MODY2 · GCK

Sequence, physicochemical and structural evidence · Human WT, disease variant and ortholog

▼ Amino-acid colour key

Colour indicates physicochemical group. Charge and relative side-chain size are listed separately.

Aliphatic · Aromatic · Polar uncharged · Positive/basic · Negative/acidic · Special

Amino acid	3-letter	1-letter	Group	Charge	Size
Alanine	Ala	**A**	Aliphatic	Neutral	Small
Valine	Val	**V**	Aliphatic	Neutral	Medium
Isoleucine	Ile	**I**	Aliphatic	Neutral	Large
Leucine	Leu	**L**	Aliphatic	Neutral	Large
Methionine	Met	**M**	Aliphatic	Neutral	Large
Phenylalanine	Phe	**F**	Aromatic	Neutral	Large
Tryptophan	Trp	**W**	Aromatic	Neutral	Large
Tyrosine	Tyr	**Y**	Aromatic	Neutral	Large
Serine	Ser	**S**	Polar uncharged	Neutral	Small
Threonine	Thr	**T**	Polar uncharged	Neutral	Medium
Asparagine	Asn	**N**	Polar uncharged	Neutral	Medium
Glutamine	Gln	**Q**	Polar uncharged	Neutral	Large
Lysine	Lys	**K**	Positive/basic	Positive	Large
Arginine	Arg	**R**	Positive/basic	Positive	Large
Histidine	His	**H**	Positive/basic	Variable (often neutral)	Medium
Aspartate	Asp	**D**	Negative/acidic	Negative	Small
Glutamate	Glu	**E**	Negative/acidic	Negative	Medium
Cysteine	Cys	**C**	Special	Variable	Medium
Glycine	Gly	**G**	Special	Neutral	Very small
Proline	Pro	**P**	Special	Neutral	Medium

1 · Evolutionary comparison

How can the human protein and its ortholog preserve a similar structure even when their sequences differ?

Sequence identity (aligned positions)
80.6%

Same physicochemical class (including identical residues)
88.4%

Global RMSD
8.18 Å

Examine the alignment and the corresponding structural views. The percentages exclude alignment gaps.

How to read the alignment

Human WT: *Homo sapiens* **Ortholog:** *Xenopus laevis*

Human SS / Ortholog SS
Secondary structure at each aligned position: α alpha helix, β beta strand, - loop or other structure.

AA Properties
* identical amino acids; : different amino acids from the same physicochemical group; × amino acids from different physicochemical groups.

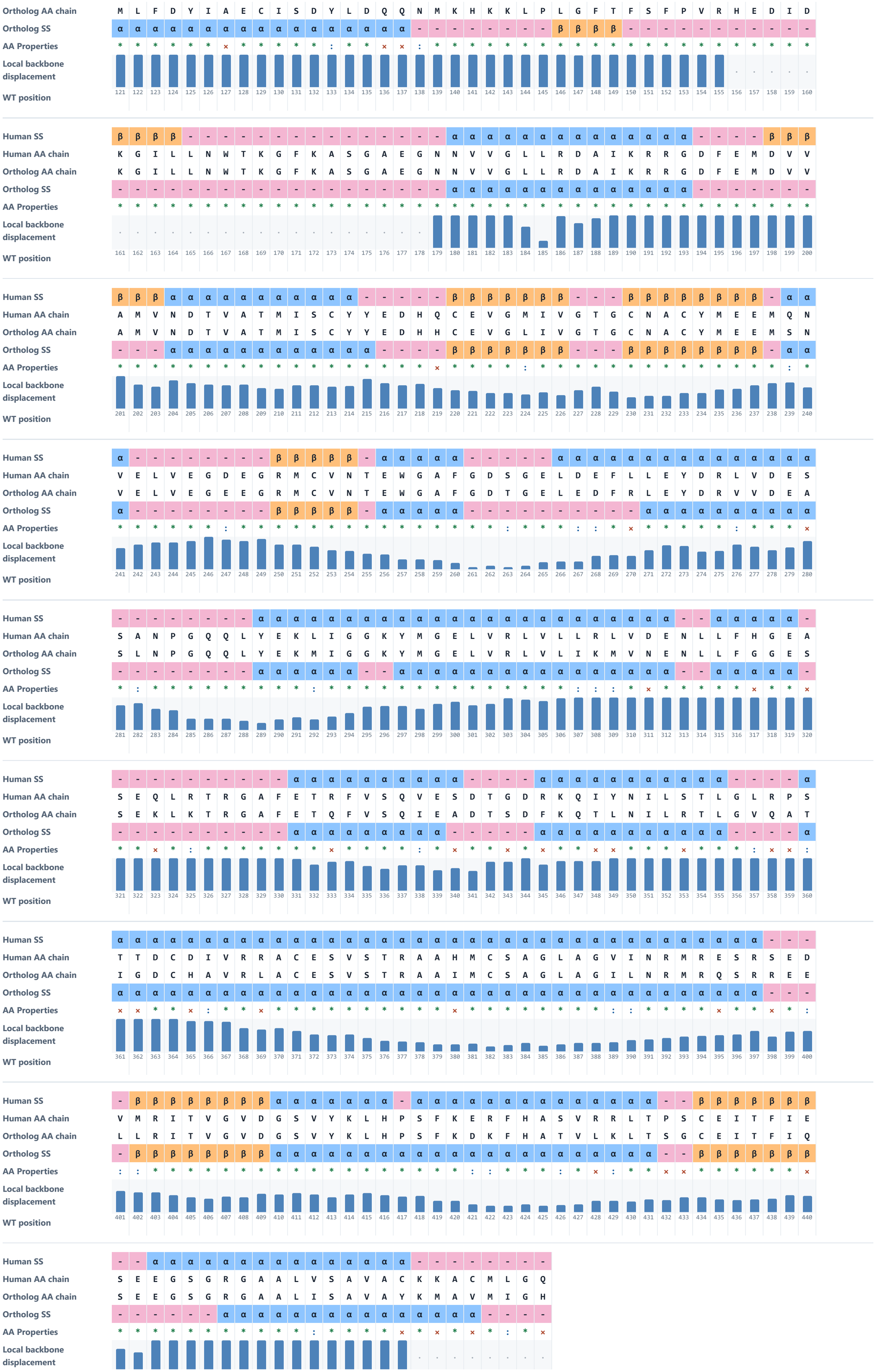
WT position
Residue number in the human WT reference sequence.

Human AA chain / Ortholog AA chain
Aligned amino-acid sequences of the human reference protein and its non-human ortholog.

Local backbone displacement
Distance between corresponding backbone α-carbons after structural superposition. Taller bars indicate greater local displacement.

α helix · β strand · - loop / other · · gap / unavailable

Local backbone displacement: bar height identifies how displaced the backbone is for each human/ortholog superposed residue. Keep in mind that loops are very flexible and can have a high displacement; this is normal. Hover for exact values.

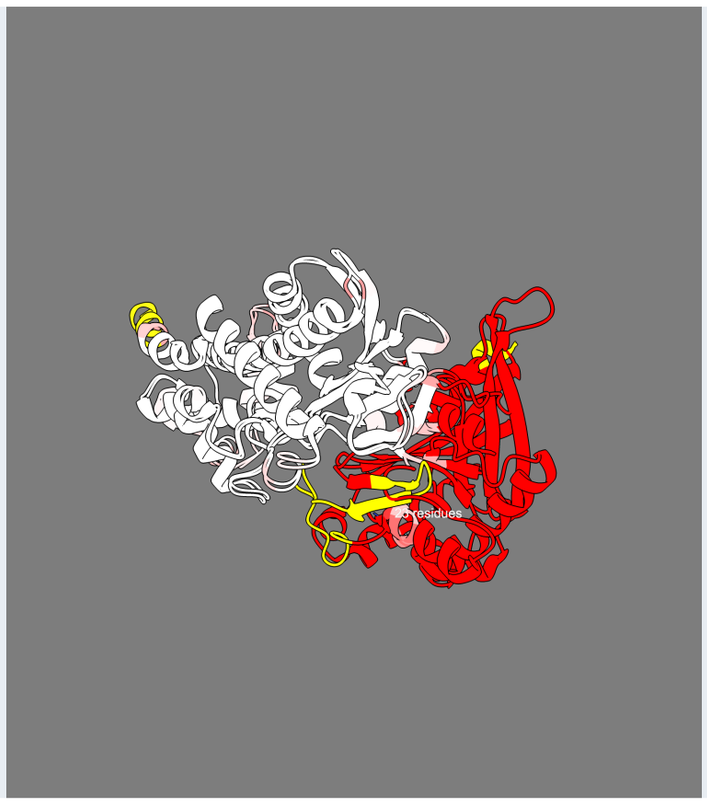

Ortholog AA chain
Ortholog SS
AA Properties
Local backbone displacement
WT position
Human SS
Human AA chain
Ortholog AA chain
Ortholog SS
AA Properties
Local backbone displacement
WT position
Human SS
Human AA chain
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Local backbone displacement
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Human SS
Human AA chain
Ortholog AA chain
Ortholog SS
AA Properties
Local backbone displacement
WT position

WT position

441 442 443 444 445 446 447 448 449 450 451 452 453 454 455 456 457 458 459 460 461 462 463 464 465

Backbone-displacement coloring

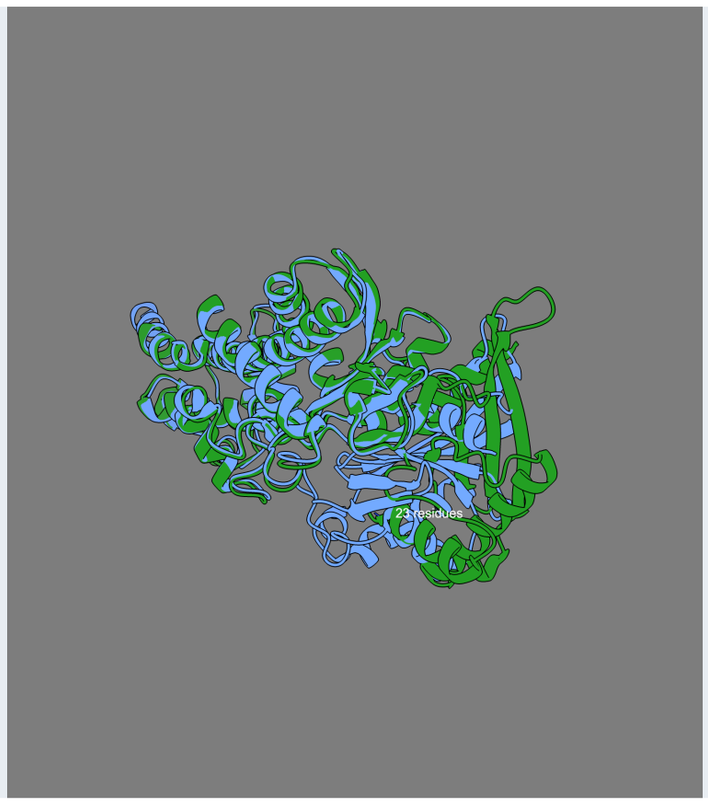
Photo | Video



Superposition of Homo sapiens and Xenopus laevis structures. They are coloured by RMSD: white means they are perfectly superposed and red means they are separated in space and badly aligned. Yellow segments identify regions that were ignored during superposition.

Human versus ortholog coloring

Photo | Video



The human-to-ortholog superposition displays Homo sapiens in blue and Xenopus laevis in green. Identify if any regions are similar in structure but appear separated in space. Sometimes, flexible loops can appear in different positions. This doesn't mean that the structures should be different; they might just have been displaced through a loop that acts as a 'hinge'.

2 · Disease-associated variant

What changes chemically at the disease-associated position, and how much does the backbone move?

Human reference position **339**

Human WT **E** · Ortholog **E** (identical)

Human WT structure

E

Glutamate 339

→

Human mutant structure

K

Lysine 339

Your comparison: use the amino-acid reference above to identify which physicochemical properties differ between the WT and mutant residues.

Local backbone displacement at variant

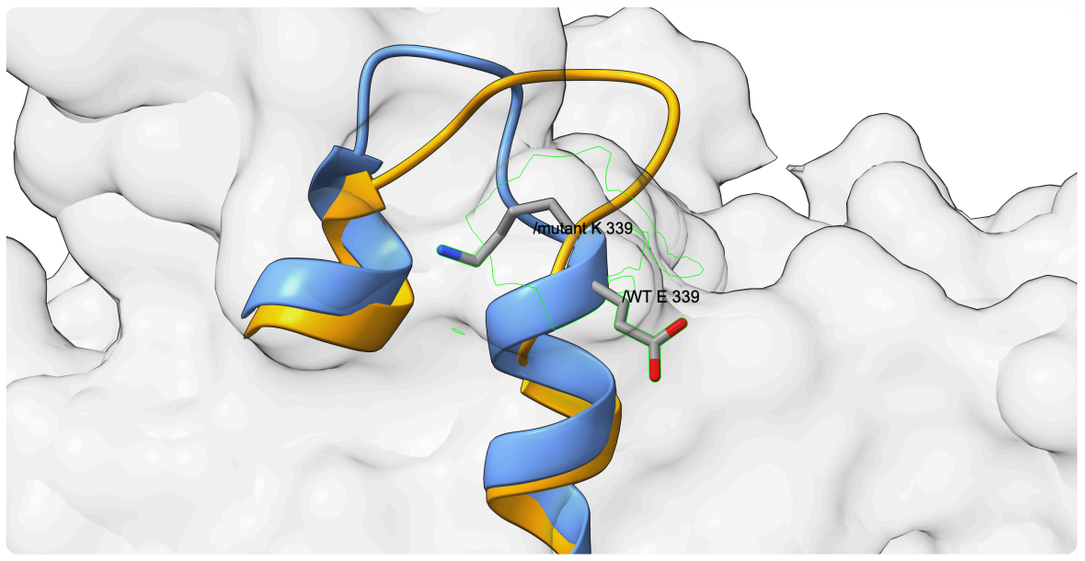
2.494 Å

Global WT–mutant RMSD: 1.483 Å. Human–ortholog local backbone displacement at the same aligned position: 3.126 Å.

Keep in mind: a small local backbone displacement can still have consequences: changes can be either structural (the protein backbone is distorted), chemical (the backbone stays the same but the amino acid chemical properties change and this impacts the interactions) or both.

WT versus mutant structural view

Photo | Video



Compare the side-chain environment with the amino-acid reference and your WT-versus-mutant comparison.